

JOINT ENTRANCE EXAMINATION (JEE ADVANCED)

Paper / Shift ID: Paper 1

Duration: 3 Hours

Maximum Marks: 300

Instructions to Candidates:

- 1. This paper contains sections for Physics, Chemistry, and Mathematics.*
- 2. Candidates must follow the instructions given per question type.*
- 3. MCQ questions have positive and negative markings as indicated.*
- 4. Numerical and Integer value questions do not have negative marking.*

Section 1: Physics

Q1. [MCQ-multiple] [+4, -1]

A block of mass $m = 2 \text{ kg}$ is placed on a rough inclined plane making an angle of 30° with the horizontal. If the coefficient of static friction is 0.5 , what is the friction force acting on the block?

- (A) 9.8 N
- (B) 4.9 N
- (C) 8.5 N
- (D) 19.6 N

Q2. [Integer] [+4, 0]

A parallel plate capacitor has plate area A and separation d . A dielectric slab of constant $K = 4$ is inserted to fill half the volume. Find the equivalent capacitance in terms of C_0 .

Section 2: Chemistry

Q1. [MCQ-multiple] [+4, -1]

Which of the following organic molecules exhibits geometric isomerism due to restricted rotation about a carbon-carbon double bond?

- (A) But-2-ene
- (B) Propene
- (C) But-1-ene
- (D) 2-Methylpropene

Q2. [Integer] [+4, 0]

For a first order reaction, the half-life is 1386 seconds. Calculate the rate constant of the reaction in s^{-1} multiplied by 10^4 .

Section 3: Mathematics

Q1. [MCQ-multiple] [+4, -1]

Let the function $f(x) = x^3 - 3x + 2$. Find the total number of local extrema of $f(x)$ on the real line.

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q2. [Integer] [+4, 0]

Evaluate the definite integral of $x \cdot \exp(x)$ from 0 to 1. Round to the nearest integer.